

Grade 11 Mathematics

Equations and Inequalities & Exponents and Surds & Trigonometry (reduction formulae, trig equations & general solutions)

Task: Task 9

Term: Term 3

Time: 60 minutes

Total Marks: 50

Examiner: AE Killer

Moderator: D Piters

QUESTION 1

1.1 Solve for x : $2x^2 - 5x - 1 = 0$ (correct to TWO decimal places) (3)

1.2 Solve for x and y simultaneously: $2x + y = 5$ and $x^2 - xy + y^2 = 7$ (6)

1.3 Determine the values of k for which the equation $x^2 + 3 = k(x - 1)$ has non-real roots. (3)

1.4 Consider the equation $(x - 5)(x + 2) = 0$.

1.4.1 Write down the values of x . (1)

1.4.2 Hence, solve for x if $(x^2 - 5)(x^2 + 2) = 0$. (1)

1.4.3 Solve for x , leaving your answer in simplest surd form:
 $(x^2 - 2x)^2 - 3(x^2 - 2x) - 10 = 0$ (3)

1.5 Solve for x : $20x \geq 4x^2 + 25$ (3)

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QUESTION 2

2.1 Solve for x : $3^{x-1} = 81$ (3)

2.2 Simplify fully, leaving your answer in expanded form: $\frac{(x+y)^{\frac{3}{2}} \cdot \sqrt{x^2 - y^2}}{(x-y)^{\frac{1}{2}}}$,
 where $x > y > 0$ (3)

2.3 Solve for x , without using a calculator: $2^{x+2} - 2^x + 2^{x-1} = 224$ (4)

2.4 Consider the expression $\sqrt[3]{\frac{1}{x^2}}$, where $x \neq 0$.

2.4.1 Write $\sqrt[3]{\frac{1}{x^2}}$ as a single power of x . (1)

2.4.2 Hence, solve for x : $\sqrt[3]{\frac{1}{x^2}} = \frac{9}{\sqrt[3]{x^4}}$ (2)

2.5 Determine the values of x for which each of the following expressions is real.

2.5.1 $\sqrt{3x-6}$ (1)

2.5.2 $\frac{\sqrt{x+4}}{x-1}$ (1)

[15]

QUESTION 3

3.1 In the diagram, $P(-8; 15)$ is a point in the Cartesian plane such that $OP = 17$ units. The reflex angle θ is measured anticlockwise from the positive x -axis to the line OP . Without using a calculator, write down the value of:

3.1.1 $\sin \theta$ (1)

3.1.2 $\tan \theta$ (1)

3.2 Prove that: $\frac{1}{\cos x} - \cos x = \sin x \cdot \tan x$ (4)

3.3 Without using a calculator, determine all values of x such that (2)

$$\sin x = \cos 35^\circ \text{ for } x \in [0^\circ; 360^\circ].$$

3.4 Answer the following without the use of a calculator.

3.4.1 Write $\sin(180^\circ + \theta)$ in terms of $\sin \theta$. (1)

3.4.2 Determine the value of $\tan 225^\circ$. (1)

3.4.3 Determine the value of $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 178^\circ + \cos 179^\circ$. (2)

3.5 Consider the expression $\frac{1 - \cos^2 x}{\tan x (\sin x - 1)}$. Determine all the values of x in the interval $x \in [0^\circ; 180^\circ]$ for which the expression is undefined. (3)

[15]

TOTAL [50]

MARKING GUIDELINE

QUESTION 1		
1.1	$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-1)}}{2(2)}$	✓ substitution
	$x = \frac{5 \pm \sqrt{33}}{4}$	
	$x = 2,69$ or $x = -0,19$	✓ ✓ both answers (3)
1.2	$y = 5 - 2x$	✓ y as subject
	$x^2 - x(5 - 2x) + (5 - 2x)^2 = 7$	✓ substitution
	$7x^2 - 25x + 18 = 0$	✓ standard form
	$(7x - 18)(x - 1) = 0$	✓ factors
	$x = 1$ or $x = \frac{18}{7}$	✓ both x- values
	$x = 1$ and $y = 3$, or $x = \frac{18}{7}$ and $y = \frac{-1}{7}$	✓ both y- values (6)
1.3	$x^2 - kx + k + 3 = 0$ so $\Delta = (-k)^2 - 4(1)(k + 3)$	✓ standard form and substitution
	$k^2 - 4k - 12 < 0$, i.e. $(k - 6)(k + 2) < 0$	✓ condition $\Delta < 0$, factorised
	$-2 < k < 6$	✓ answer (3)

1.4.1	$x=5$ or $x=-2$	✓ both answers (1)
1.4.2	$x^2=5$ (reject $x^2=-2$), so $x=\pm\sqrt{5}$	✓ answer with rejection (1)
1.4.3	Let $k=x^2-2x$: $k^2-3k-10=0$	✓ substitution
	$(k-5)(k+2)=0$, so $x^2-2x=5$ or $x^2-2x=-2$	✓ factors and back-substitution
	$x^2-2x+2=0$ has no real roots ($\Delta=-4<0$); $x=1\pm\sqrt{6}$	✓ answer with rejection (3)
1.5	$4x^2-20x+25\leq 0$	✓ standard form
	$(2x-5)^2\leq 0$	✓ factors
	$x=\frac{5}{2}$	✓ answer (3)

QUESTION 2		
2.1	$3^{x-1}=3^4$	✓ same base
	$x-1=4$	✓ equate exponents

	$x=5$	✓ answer (3)
2.2	$= \frac{(x+y)^{\frac{3}{2}} \cdot \sqrt{(x+y)(x-y)}}{(x-y)^{\frac{1}{2}}}$	
	$= \frac{(x+y)^{\frac{3}{2}} \cdot (x+y)^{\frac{1}{2}} \cdot (x-y)^{\frac{1}{2}}}{(x-y)^{\frac{1}{2}}}$	✓ factorise and rewrite surd as fractional exponents
	$= (x+y)^2$	✓ combine exponents
	$x^2 + 2xy + y^2$	✓ expanded answer (3)
2.3	$2^{x-1}(2^3 - 2^1 + 1) = 224$	✓ factorise lowest common power
	$7 \cdot 2^{x-1} = 224 \text{ so } 2^{x-1} = 32$	✓ simplify coefficient
	$2^{x-1} = 2^5$	✓ same base
	$x=6$	✓ answer (4)
2.4.1	$\sqrt[3]{\frac{1}{x^2}} = x^{\frac{-2}{3}}$	✓ single negative fractional exponent (1)

2.4.2	$x^{-\frac{2}{3}} = 9x^{-\frac{4}{3}}$, so $x^{\frac{2}{3}} = 9$	✓ single fractional exponent
	$x = \pm 9^{\frac{3}{2}}$, so $x = \pm 27$	✓ reciprocal power and both answers (2)
2.5.1	$3x - 6 \geq 0$ so $x \geq 2$	✓ radicand non-negative (1)
2.5.2	$x \geq -4$ and $x \neq 1$	✓ both conditions (1)

QUESTION 3		
3.1.1	$\sin \theta = \frac{15}{17}$	✓ answer (1)
3.1.2	$\tan \theta = \frac{-15}{8}$	✓ answer (1)
3.2	$\text{LHS} = \frac{1 - \cos^2 x}{\cos x}$	✓ common denominator
	$= \frac{\sin^2 x}{\cos x}$	✓ square identity
	$= \sin x \cdot \frac{\sin x}{\cos x}$	✓ split factors
	$= \sin x \cdot \tan x = \text{RHS}$, so $\frac{1}{\cos x} - \cos x = \sin x \cdot \tan x$ (LHS = RHS)	✓ quotient identity to

		RHS (4)
3.3	$\sin x = \sin(90^\circ - 35^\circ) = \sin 55^\circ$	✓ co-function conversion
	$x = 55^\circ$ or $x = 125^\circ$	✓ both answers (2)
3.4.1	$\sin(180^\circ + \theta) = -\sin \theta$	✓ reduction (1)
3.4.2	$\tan 225^\circ = \tan 45^\circ = 1$	✓ reduction and answer (1)
3.4.3	$\cos(180^\circ - \theta) = -\cos \theta$, so $\cos 1^\circ + \cos 179^\circ = 0$, $\cos 2^\circ + \cos 178^\circ = 0$, ... , $\cos 89^\circ + \cos 91^\circ = 0$	✓ pairing using reduction
	$\cos 90^\circ = 0$, so the sum is 0	✓ answer (2)
3.5	$\tan x = 0$ gives $x = 0^\circ$	✓ value from $\tan x = 0$
	$\sin x = 1$ or $\tan x$ undefined ($\cos x = 0$) gives $x = 90^\circ$	✓ value from $\sin x = 1$ or $\cos x = 0$
	$x = 0^\circ$, $x = 90^\circ$ or $x = 180^\circ$	✓ value 180° and

		final answer (3)

TOTAL [50]**COGNITIVE LEVEL ANALYSIS GRID**

Weighting (Prescribed)			20%			35%			30%			15%			100%
Actual %			20%			34%			32%			14%			100%
Topic	Q	Sub-topic	Knowledge			Routine			Complex			Problem Solving			Total
			E	M	D	E	M	D	E	M	D	E	M	D	
Equations and Inequalities	1.1	quadratic formula to 2 d.p.		3											3
	1.2	linear-quadratic simultaneous equations					6								6
	1.3	non-real roots, parameter range								3					3
	1.4.1	roots from factorised form	1												1
	1.4.2	biquadratic via substitution				1									1
	1.4.3	substitute repeated expression, solve												3	3
	1.5	perfect square quadratic inequality							3						3
Exponents and Surds	2.1	exponential eq, common base	3												3
	2.2	simplify surd with fractional exponents					3								3
	2.3	exponential eq, factor common power								4					4
	2.4.1	root as fractional exponent				1									1
	2.4.2	solve equation, reciprocal power												2	2
	2.5.1	surd real, radicand non-negative				1									1
	2.5.2	surd and denominator								1					1

		restrictions													
Trigonometry (reduction formulae, trig equations & general solutions)	3.1.1	sine from point, definition	1												1
	3.1.2	tangent from point, definition	1												1
	3.2	prove trigonometric identity					4								4
	3.3	co-function trig equation, interval								2					2
	3.4.1	recall reduction formula	1												1
	3.4.2	special angle via reduction				1									1
	3.4.3	sum of cosines by pairing												2	2
	3.5	values where trig expression undefined									3				3
TOTAL			7	3	0	4	13	0	3	6	7	0	0	7	50
%			20%			34%			32%			14%			100%

E = Easy, M = Medium, D = Difficult.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1+i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x-a)^2 + (y-b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$